

Subject: / /

Summarize :-

- (i) Define a state space that contain all possible cond. of the relevant objects.
- (ii) Specify one or more state that describe possible situation from which the problem solving may start (i.e. initial state)
- (iii) Specify one or more state that will be accepted as goal state.
- (iv) Specify a set of rules that describe the actions available.

Production System →

→ Set of rule, left side specify the applicability of the rule and right side specify operation to be performed.

→ One or more knowledge database

→ **Control strategy** :- specify the order in which rules will be compared in the database.

→ A way of resolving conflicts when several rules match at once.

→ Rule applicator

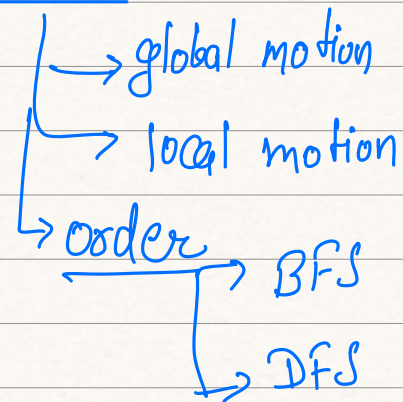
Subject: * Control strategies

* what next rule i should apply so that my goal state is achieved.

→ which rule to apply next for achieving the goals.

(i) It should cause motion

(ii) Control strategy must be systematic



BFS

↳

- create a variable NODE_LIST and set it to initial state

- Until a Goal state is found or NODE_LIST is empty do

(a) Remove the first element from NODE_LIST and call it 'E'.
NODE_LIST was empty then QUIT

(b) For each way the rule can match the state described in 'E'

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- ↳ (i) apply the rule / /
to generate a new state
- (ii) If new state is goal state
then QUIT
- (iii) If new state is not goal
state add this state to
NODE-LIST

BFS advantages

↳ (i) BFS will not get trapped exploring
a blind alley

(ii) If there is a solution, BFS guarantees you
to give the solⁿ.

Furthermore if there are
multiple solⁿ, you will get the minimal
solⁿ.

DFS :-

1. If initial state is GOAL state then QUIT.

2. Do the following UNTIL SUCCESS or FAILURE is
designated:-

(a) Generate a successor 'f' of the initial
state. If no successor then QUIT.

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(b) Call depth first search with E as initial state.

(c) If success is returned signal success otherwise continue in loop.

Advantages:-

- (i) less m/m.
- (ii) DFS may find the solⁿ without exploring the much of the state space.

To Do

→ Travelling Salesman problem apply this topic